

Worksheet

Single event probabilities

- 1 The diameters (in millimetres) of nectarines in a bag of nectarines picked at random at a Mildura orchard are recorded below.

60 55 54 49 49 57 55 55 58 64 65 47 52
56 56 59 62 60 45 46 49 53 53 58 57

- a Nectarines with a diameter less than 50 mm are classified as small. Find the probability of picking a small nectarine.

- b Nectarines with a diameter greater than 60 mm are classified as large. Find the probability of picking a large nectarine.

- c All other nectarines are classified as regular. Find the probability of picking a regular nectarine?

- d If the nectarine crop was 10 000 nectarines, how many nectarines would you expect to be large.

- 2 A stationery shop has 120 blue pens, 80 black pens and 50 red pens on the clearance table. If a pen is selected at random find the probability that it is:

a blue

b red

c black

d not red.

- 3 Belinda plays basketball in the local competition and recorded her scores in the summer season. Her scores were:

12 6 8 16 15 24 30 15 18 2 5
15 10 11 12 6 21 20 4 14 12 16

- a Find the probability that Belinda scores:

i more than 10 points in a game

ii at least 20 points in a game

iii at most, 10 points in a game.

- b If Belinda played 40 games in a year, in how many games would you expect her to score less than 10 points?

- 4 Robert has four different shirts. They are white, blue, striped and brown. He also has three ties that are red, black and pastel.

a Show the sample space of combinations of shirts with ties.

b Find the probability that Robert wears a white shirt with a red tie.

c Find the probability that Robert wears a white shirt with a red tie or a black tie.

d Find the probability that Robert wears a striped shirt and wears a tie that is not red.

e Find the probability that Robert wears a shirt that is not white with a tie that is not pastel.

- 5 For her bicycle lock, Addie had to choose a secret code that is a letter and a single digit.

a Find the size of the sample space of combinations of letters and numbers that Addie could have chosen as her code.

b Find the probability that Addie chose the code E3.

c Find the probability that Addie chose a code with the letter E and a number greater than 3.

d Find the probability that Addie chose a code with the letters E or U and the numbers 3 or 5

e Find the probability that Addie chose a code that does not contain the letter E or the number 3

- 6 A letter is selected at random from the sentence: 'EVERY PICTURE TELLS A STORY.' Find the probability the letter selected is:

a a vowel

b a consonant

c a 'T' or a 'Y'

d not the letter 'T'

e a letter in the word 'PICTURE'.

Answers

1 a $\frac{7}{25}$ b $\frac{3}{25}$ c $\frac{3}{5}$ d 1200

2 a $\frac{12}{25}$ b $\frac{1}{5}$ c $\frac{8}{25}$ d $\frac{4}{5}$

3 a i $\frac{15}{22}$ ii $\frac{2}{11}$ iii $\frac{7}{22}$

b 11

4 a {(white, red), (white, black), (white, pastel), (blue, red), (blue, black), (blue, pastel), (striped, red), (striped, black), (striped, pastel), (brown, red), (brown, black), (brown, pastel)}

b $\frac{1}{12}$ c $\frac{1}{6}$ d $\frac{1}{4}$ e $\frac{1}{2}$

5 a 260 b $\frac{1}{260}$ c $\frac{3}{130}$ d $\frac{1}{65}$ e $\frac{45}{52}$

6 a $\frac{8}{23}$ b $\frac{15}{23}$ c $\frac{5}{23}$ d $\frac{20}{23}$ e $\frac{14}{23}$